

Software Requirements Specification (SRS)

Car Parts ERP + POS + Ecommerce Management System

1. Introduction

1.1 Purpose

The purpose of this document is to define the complete Software Requirements Specification (SRS) for the Car Parts ERP + POS + Ecommerce System. This system will manage online sales, physical shop billing (POS), service invoicing, accounting, HRM, inventory, and reporting for a car parts business.

1.2 Scope

The system will be a web-based application that supports:

- Ecommerce for car parts
- POS for physical shop
- Sales & Service Invoice
- Inventory Management
- Accounting (Basic)
- HRM (Basic for 10–15 employees)
- Reporting & Analytics

The system will be used by administrators, sales staff, accountants, and customers.

1.3 Definitions

- POS: Point of Sale (Physical shop billing system)
- ERP: Enterprise Resource Planning
- SKU: Stock Keeping Unit
- HRM: Human Resource Management

2. Overall Description

2.1 Product Perspective

This system is an integrated business management platform for a car parts shop that supports both online and offline operations in a single system.

2.2 Product Features (High-Level)

- Ecommerce website
- Physical shop POS billing
- Service & sales invoice creation
- Manual discount system
- Inventory & low stock alert
- Accounting module
- HRM (basic)
- Monthly revenue dashboard

2.3 User Classes & Roles

1. Super Admin – Full system control
2. Admin – Manage products, orders, employees
3. Sales Staff – POS & invoice management
4. Accountant – Accounting & reports
5. Customer – Ecommerce user

2.4 Operating Environment

- Web-based application (Browser)
- Mobile & Desktop responsive
- Server: Cloud/VPS/cPanel hosting
- Database: MySQL or equivalent

3. Functional Requirements

3.1 Ecommerce Module

3.1.1 Product Management

- Add/Edit/Delete products
- Product name, SKU, category, brand
- Multiple product images upload
- Price and stock management
- Product description & specifications

3.1.2 Category & Brand Management

- Create categories (Engine Parts, Brake, Oil, etc.)
- Create brands
- Assign category & brand to products

3.1.3 Customer Features

- Customer registration & login
- Product browsing & search
- Add to cart
- Checkout system
- Order history & tracking

3.1.4 Order Management

- Online order listing
 - Order status (Pending, Processing, Delivered, Cancelled)
 - Payment method selection (COD, bKash, etc.)
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3.2 POS (Physical Order System)

3.2.1 POS Billing Screen

- Fast product search (Name/SKU/Barcode)
- Add product to cart
- Quantity update
- Instant total calculation
- Walk-in customer billing

3.2.2 Payment Options

- Cash
- Mobile Payment (bKash/Nagad)
- Card
- Partial/Mixed payment

3.2.3 Invoice Print

- Printable invoice (A4 & POS format)
 - Auto invoice number generation
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3.3 Sales & Service Invoice Module

3.3.1 Sales Invoice

- Customer selection
- Multiple product addition
- Discount (Fixed & Percentage)
- Tax (optional)
- Grand total auto calculation

3.3.2 Service Invoice

- Service type selection
 - Service charge input
 - Parts used (optional)
 - Technician name
 - Service notes
 - Printable service invoice
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3.4 Manual Discount System

- Per product discount
 - Overall invoice discount
 - Fixed amount discount
 - Percentage discount
 - Discount permission control (Admin only)
 - Discount history tracking
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3.5 Custom Product Billing (Out of Ecommerce)

- Manual product entry during billing
 - Temporary item name & price input
 - Non-listed product billing support
 - Quick billing feature
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3.6 Inventory Management

3.6.1 Stock Management

- Real-time stock tracking
- Stock in & stock out history
- Supplier restock tracking

3.6.2 Low Stock Alert

- Minimum stock threshold setting
 - Dashboard low stock alert
 - Notification for low stock items
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3.7 Accounting Module (Basic)

3.7.1 Income & Expense

- Sales income tracking
- Expense entry (manual)
- Expense categories

3.7.2 Financial Reports

- Profit & Loss report
- Daily sales report
- Monthly financial summary
- Cashbook/Ledger

3.7.3 Due & Payment Tracking

- Customer due tracking

- Supplier payment tracking
 - Payment history
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3.8 HRM Module (Basic – 10–15 Employees)

3.8.1 Employee Management

- Employee profile (Name, Role, Contact)
- Role & permission management
- Employee status (Active/Inactive)

3.8.2 Attendance & Salary

- Basic attendance tracking
 - Salary management
 - Salary history records
 - Leave management (basic)
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3.9 Reports & Analytics Dashboard

- Monthly revenue report
- Daily sales report
- Top selling products
- Online vs Offline sales report
- Profit summary
- Export reports (PDF/Excel)

Dashboard Widgets:

- Total Monthly Revenue
 - Total Orders
 - Low Stock Items
 - Total Profit
 - Pending Orders
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4. Non-Functional Requirements

4.1 Performance Requirements

- System should support 50+ concurrent users
- POS billing response time < 2 seconds
- Fast product search capability

4.2 Security Requirements

- Role-based authentication
- Secure login system
- Data encryption for sensitive data
- Admin access control

4.3 Usability Requirements

- User-friendly dashboard UI
- Responsive design (Mobile + Desktop)
- Easy POS interface for quick billing

4.4 Reliability

- Daily database backup
- Error logging system
- High uptime (99%)

4.5 Scalability

- Future multi-branch support
- API-ready architecture
- Modular system design

5. System Architecture (Recommended)

- Stack: Laravel Inertia js React
- Database: MySQL
- Hosting: VPS or Cloud Server
- Payment Integration: bKash / Online Gateway

6. Future Enhancements (Optional)

- Barcode scanning system
 - Multi-branch management
 - Supplier management module
 - Warranty tracking for car parts
 - Mobile app integration
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7. Acceptance Criteria

The system will be considered complete when:

- E-commerce, POS, Invoice, Accounting, HRM modules are fully functional
 - Reports & dashboard show accurate real-time data
 - Low stock alerts work correctly
 - Invoices and reports are printable & exportable
 - Role-based access control is implemented
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8. Estimated Modules Summary

1. E-commerce System
2. POS & Billing System
3. Invoice & Service Module
4. Inventory Management
5. Accounting (Basic)
6. HRM (Basic)
7. Reports & Analytics Dashboard

End of Document.